Epistemic Verification of Physics Claims via Hyperdimensional Structural Search

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Abstract

We present an epistemic verification engine for scientific claims that combines Hyperdimensional Computing (HDC) structural search with a three-dimensional epistemic classification framework (LEM Cube). Our system encodes 208 physics equations across 20 domains as 16,384-dimensional hypervectors, enabling structural analog discovery via skeleton encoding—a technique that strips variable names to reveal pure mathematical structure. We demonstrate the system on 10 high-profile case studies organized in 5 paired comparisons, including the cold fusion/NIF ignition contrast, the LK-99/nickelate superconductor divergence, and the Schön/Dias fraud pattern. The system correctly classifies verified discoveries (LIGO, Higgs) as E4/N3/M3, debunked claims (cold fusion, EMDrive) as E0/N0/M0, and active controversies (Hubble tension) as E4/N2/M3. We additionally report a nuclear discovery engine calibrated against 70 AME2020 reference nuclei that predicts a superheavy stability island centered at Z=115-120, N=180, consistent with published theoretical predictions. The complete system—including physics catalog, discovery bridge, nuclear engine, and decentralized science portal—is open source under AGPL-3.0.

1 Introduction

The reproducibility crisis in science is well-documented: up to 70% of biomedical studies fail replication [Ioannidis, 2005], fraudulent publications are outpacing legitimate growth [Nature, 2025], and 2.6% of 2025 papers contain AI-hallucinated citations [Nature, 2026]. Existing Decentralized Science (DeSci) platforms address publishing and funding but lack computational verification of claims against established physics [DeSci, 2025].

We present three contributions:

1. A **physics equation catalog** of 208 landmark equations encoded as 16,384-dimensional continuous hypervectors (ContinuousHV) using Hyperdimensional Computing (HDC), enabling structural analog discovery across 20 physics domains.
2. A **discovery bridge** that classifies physics predictions via the LEM Cube—a three-dimensional epistemic framework with Empirical (E0–E4), Normative (N0–N3), and Materiality (M0–M3) axes.
3. A **nuclear discovery engine** calibrated against 70 AME2020 reference nuclei, predicting superheavy element stability via Bethe-Weizsäcker SEMF + Woods-Saxon shell model.

2 Method

2.1 Hyperdimensional Equation Encoding

Each equation is encoded via four parallel channels:

1. **Full encoding**: Recursive AST traversal with role-binding. Each node type (Field, Constant, DiffOperator, Sum, Product, Exponential, Power) gets a deterministic role vector. Children are position-bound to preserve ordering.
2. **Skeleton encoding**: Names stripped, leaving pure operator structure. This enables cross-domain analogy: the Helmholtz and Schrödinger equations have *identical skeletons* ($\nabla^2\psi + k^2\psi = 0$) despite different physics domains.
3. **Symmetry encoding**: Lie groups (SO(n), SU(n), U(1), Lorentz, Poincaré) and discrete symmetries (C, P, T, CPT, $\mathbb{Z}_2$) encoded via algebra dimension scaling.
4. **Dimensional encoding**: 7 SI base dimensions as orthogonal basis vectors, scaled by exponent.

Similarity is computed as a weighted combination:

$$S = 0.4 \cdot S_{\text{skeleton}} + 0.3 \cdot S_{\text{symmetry}} + 0.2 \cdot S_{\text{dimensional}} + 0.1 \cdot S_{\text{full}} \tag{1}$$

2.2 LEM Cube Classification

The LEM Cube assigns three orthogonal epistemic coordinates:

- **E-axis (Empirical)**: E0 (unverifiable) through E4 (publicly reproducible), determined by simulation confidence and data availability.
- **N-axis (Normative)**: N0 (personal) through N3 (axiomatic), determined by catalog similarity—high similarity to known equations implies higher normative authority.
- **M-axis (Materiality)**: M0 (ephemeral) through M3 (foundational), determined by the physics domain.

3 Catalog

The catalog contains 208 equations across 20 physics domains (Table 1).

4 Case Studies

We demonstrate the system on 10 case studies in 5 paired comparisons (Table 2). Each pair contrasts a verification failure with a verification success using the same physics.

4.1 Structural Search Results

For the Lazar "Gravity-A" claim, the top-5 structural neighbors are: Heat Equation (0.629), Yukawa Potential (0.603), Schwarzschild Metric (0.583), de Sitter Metric (0.581), FLRW Metric (0.579). The claim sits between nuclear force and gravitational field equations at moderate similarity ($\sim$0.6), indicating a structural analog exists but no exact match.

For the Art's Parts metamaterial, the nearest neighbor is the Waveguide Dispersion Relation at 0.915 similarity—the claimed physics is textbook optics. ORNL (2022) confirmed the physical sample fails.

Table 1: Equation catalog coverage by domain.

Domain	Count	Domain	Count
Classical Mechanics	14	Optics	8
Electromagnetism	13	Condensed Matter	14
General Relativity	12	Particle Physics	6
Quantum Mechanics	17	Modified Gravity	3
Quantum Field Theory	4	Information Theory	14
Statistical Mechanics	13	Biophysics	9
Fluid Dynamics	7	Acoustics	3
Cosmology	6	Plasma Physics	11
Nuclear Physics	9	Mathematics	4
Thermodynamics	18	Astrophysics	6

5 Nuclear Discovery

The nuclear discovery engine combines Bethe-Weizsäcker SEMF with Woods-Saxon shell model corrections, calibrated against 70 AME2020 reference nuclei. A systematic sweep of Z=110-120, N=170-190 reveals a stability island centered at Z=115, N=180 with shell corrections -17 to -23 MeV, consistent with published predictions by Möller et al. and Oganessian & Utyonkov.

6 Discussion

Limitations. The SEMF + simplified shell model is educational, not research-grade; calibrated predictions carry $\sigma > 5$ MeV uncertainty for superheavy elements. The structural search uses name-hash encoding for 106 of 208 equations, limiting skeleton clustering accuracy for those entries.

Comparison. No existing DeSci platform (DeSci Labs, ResearchHub, Bio Protocol) performs computational verification of claims against a physics equation catalog. ResearchHub's AI reviewer catches math errors; our system identifies structural analogs across 20 domains.

7 Conclusion

We have demonstrated that hyperdimensional structural search, combined with multi-dimensional epistemic classification, provides a practical framework for automated scientific claim verification. The system correctly distinguishes verified discoveries, debunked claims, fraudulent publications, and active controversies. The open-source implementation at `https://desci.mycelix.net` serves as both a DeSci portal and an educational physics resource.

References

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Table 2: Case studies with LEM classifications.

Case	E	N	M	Verdict
Cold Fusion (1989)	0	0	0	Never replicated; Gamow factor $\sim 10^{-2700}$
NIF Ignition (2022–25)	4	3	3	Q: 1.54 $\rightarrow$ 4.13; 8 ignitions
LK-99 (2023)	0	0	0	Cu_2S impurity; debunked in 3 weeks
Nickelates (2023–25)	3	2	2	Multiple groups; SLAC ambient pressure
Schön (2002)	0	2	0	Identical noise; 25+ retractions
Dias (2020–23)	0	2	0	Same pattern; 5 retractions
LIGO GW150914	4	3	3	SNR=24; false alarm $< 1/203{,}000$ yr
Higgs boson (2012)	4	3	3	ATLAS+CMS; 5.9σ
Hubble Tension	4	2	3	Both sides E4; 5σ discrepancy
EMDrive	0	0	0	Conservation violation; thrust=0 shielded

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